
Introduction

Transdisciplinary education enables students, researchers, teachers, and other stakeholders to work on complex wicked problems and develop skills for an ever-changing future. Translating such ambitions into concrete educational modules is a significant challenge.

That's why it's important to explicitly consider **learning objectives and learning outcomes** when designing a transdisciplinary education project. Working with (learning) objectives and (learning) outcomes is a foundational principle in education and is closely tied to student assessment.

Click the **Theory** tab to understand the difference between process-related and product-related learning outcomes.

Via the **Practice** tab, you'll find a tool to help formulate learning outcomes and objectives for transdisciplinary education.

Theory

Transdisciplinary education focuses on developing an attitude and way of being that enables individuals to contribute to complex societal challenges (Veltman et al., 2021). This requires learning outcomes that focus on both process and product (Daneshpour & Kwegyir-Afful, 2022; Ncter, 2025).

Process

The process involves the knowledge, skills, and attitudes developed during transdisciplinary collaboration. Within universities of applied sciences and research universities — and in the literature — there are various frameworks outlining these competencies.

The list below is an example and can help formulate the knowledge, skills, and attitudes you want students to develop (e.g., Di Giulio & Defila, 2017; Oudenampsen et al., 2024; Gulikers & Oonk, 2019; Ncter, 2025).

- **Self-awareness** Awareness of the role your own discipline plays in a given issue and its limitations, as well as the ability to recognize the added value of other disciplines and non-academic contributors.
- **Purposeful collaboration** The ability to intentionally shape and organize collaboration and dialogue with stakeholders from diverse backgrounds and disciplines in a way that contributes to solving the issue.
- **Iterative opinion formation and mutual learning** The ability to engage openly in dialogue with people from different backgrounds and disciplines, including non-academics, to influence others' perspectives and allow your own to evolve.
- **Integrating different perspectives** The ability to integrate your own professional perspective and methods with

those of others (scientific, practical, and experiential sources) in collaboration with people from other disciplines and non-academics, to co-create solutions for complex societal challenges. This process is also referred to as “perspectival agility” — more on this at the bottom of the page.

Product

Solutions to complex societal challenges are rarely final. They require ongoing refinement and adaptation to changing circumstances. Therefore, when designing a transdisciplinary education project, it's important to formulate realistic expectations regarding learning outcomes in relation to the societal issue. Allow for variation in the final product — in terms of form, structure, or degree of completion.

A good final product doesn't need to be “finished,” but it must be meaningful for both the learning process and the involved practice partners.

A transdisciplinary project can yield different types of outcomes, depending on the issue and the collaboration (Losse, 2016):

- **Advice**

Providing knowledge, insights, or lines of thought that contribute to possible solution directions.

- **Design**

Developing a new idea, concept, or experiment that contributes to the issue.

- **Intervention**

A concrete solution tested in practice, such as a pilot or prototype.

Perspectival Flexibility: A Key New Area of Research

This is the term researchers use to describe the process in which individuals can recognize, interpret, and analyze different perspectives, reflect on them, and potentially adjust their own viewpoint.

These skills are essential in transdisciplinary collaborations, as the central issues are often so complex that many different perspectives and angles are needed to arrive at effective solutions.

However, this is a complex cognitive process, and research shows that adolescents are increasingly less skilled in this area.

That's why researchers at Avans University of Applied Sciences are conducting two studies aimed at

understanding the mechanisms behind this social-cognitive skill.

Taking and combining perspectives appears to consist of three subprocesses:

1. Perceiving perspectives
2. Distinguishing between one's own and others' perspectives
3. Combining perspectives

People often find it difficult to look beyond their own frame of reference, which makes it challenging to fully utilize other perspectives.

Especially when information is contradictory, it becomes hard to identify common ground — even though such commonalities are often essential for resolving conflicts or reaching shared insights.

As a result, opportunities to develop new ideas and solutions may go unused.

The two studies, conducted by the Research Group on Transdisciplinary Collaboration in Education and the Research Group on Brain & Learning, are currently underway.

Ultimately, the researchers aim to provide concrete tools for higher education, so that students are better equipped to collaborate on the societal challenges of today and

tomorrow.

Want to stay up to date on the outcomes, latest insights, and practical tools related to perspectival flexibility?

Sign up for the newsletter or follow the Avans Future-proof Education Center on LinkedIn.

Want to learn more now?

Contact Associate professor Lara van Peppen
m.vanpeppen@avans.nl Or visit: [Onderwijs met perspectief: adaptieve ondersteuning bij \(het aanleren van\) perspectivistische lenigheid | Avans Hogeschool](#)

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Practice

This tool provides insight into what education developers need to consider when formulating learning outcomes for multi-, inter-, and transdisciplinary modules. [Tool formuleren leeruitkomsten multi-, inter-, transdisciplinaire modules Avans. maart 2025.pdf](#)